

REVIEW

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Amyloid solutions: lecanemab, gantenerumab, and donanemab in the treatment of Alzheimer's disease

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Abstract

Alzheimer's disease (AD), the leading cause of dementia, is a multifaceted neurodegenerative disorder characterized by amyloid-beta (A β) aggregation, tau pathology, and neuroinflammation. Recent advances in immunotherapy have yielded monoclonal antibodies (mAbs) with unprecedented precision in targeting A β species, offering hope for altering the disease trajectory. Lecanemab's selective affinity for protofibrillar A β disrupts early neurotoxic aggregates, providing a proactive approach for mitigating synaptic dysfunction. Donanemab, by targeting pyroglutamate-modified plaques, combines high specificity with microglial-mediated clearance, while gantenerumab enhances Fc receptor-mediated phagocytosis of both soluble and fibrillar A β . These therapies represent a shift from symptomatic treatment to disease modification, reducing the A β burden and slowing cognitive decline. However, challenges, such as amyloid-related imaging abnormalities (ARIA), underscore the complexity of balancing efficacy and safety. This review explores the transformative potential of these therapies and contextualizes their mechanisms within the evolving understanding of AD pathophysiology. This highlights the interplay between immune modulation, neuroprotection, and personalized medicine, highlighting the need for future innovations in tackling one of the most formidable challenges in neurology.

Keywords Alzheimer's disease, Amyloid-beta, Monoclonal antibodies, Immunotherapy, Lecanemab, Donanemab, Gantenerumab, Neuroinflammation, Disease modification, ARIA

Introduction

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by irreversible cognitive decline, pervasive memory loss, and reduced functional autonomy. Together with Parkinson's disease (PD), these conditions dominate the landscape of age-associated neurological

disorders and contribute significantly to global morbidity. AD alone accounts for approximately 30–35 million cases worldwide, disproportionately affecting older adults and profoundly impacting their quality of life and independence. PD affects over 10 million individuals globally, and its prevalence increases sharply with advancing age. Alarming projections suggest that by 2050, AD will affect 115 million individuals, primarily targeting 10–30% of those aged 65 and older, creating immense public health and economic burdens [1–4]. This epidemic underscores the urgency of advancing research on its etiopathogenesis and novel treatment modalities.

AD is characterized by aberrant protein aggregates, including beta-amyloid (A β) and hyperphosphorylated

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tau, which disrupt synaptic function, induce neuroinflammation, and lead to neuronal loss. Neuropsychological impairments correlate strongly with disease progression and cognitive decline [5]. Concomitant neuroanatomical alterations, including hippocampal atrophy and A β plaque deposition, underscore complexity of AD in patients. Emerging research highlights the role of adaptive immune mechanisms in modulating A β deposition and tau misfolding, offering a new perspective on AD pathophysiology [6, 7]. Immunomodulatory approaches are being explored to prevent or mitigate A β and tau accumulation, emphasizing adaptive immunity's role in regulating these neurotoxic protein aggregates [8]. Monoclonal antibodies, such as lecanemab, gantenerumab, and donanemab, represent groundbreaking approaches for AD treatment. Their development marks a significant shift toward precision medicine in AD, offering hope for the more effective management of this devastating disorder. This underscores the importance of advancing research on amyloid-targeting therapies and their integration into clinical practice.

Figure 1 outlines the cascade initiated by A β accumulation, highlighting the role of microglia and astrocytes in inflammation and subsequent neurodegeneration in AD.

Monoclonal antibodies (mAbs) that target A β have emerged as a promising class of disease-modifying agents. These therapies demonstrate unique epitope selectivity, enabling the clearance of distinct A β conformations, including insoluble plaques and soluble oligomers, which are pivotal for AD pathogenesis [8]. Recent breakthroughs include mAbs, such as lecanemab, donanemab, and remternetug, each with distinct binding profiles and mechanisms. Lecanemab, a recently granted regulatory approval, preferentially targets protofibrillar A β , a large oligomeric species implicated in early neurotoxicity. This selectivity is critical because protofibrils are more neurotoxic than their monomeric or fibrillar counterparts. The enhanced avidity of lecanemab for protofibrils, coupled with its capacity to reduce both soluble and insoluble A β burdens, represents a paradigm shift in AD therapy [8, 9]. Unlike traditional therapies such as donepezil and galantamine, which offer symptomatic relief, lecanemab addresses underlying disease processes, marking a significant advancement in the field [10, 11].

Similarly, donanemab and remternetug target pyroglutamylated A β , a plaque-specific variant, leveraging immune effector mechanisms to clear these aggregates [8, 9]. The dual-action mechanism of these mAb—disrupting clusterization and facilitating microglial-mediated phagocytosis—has shown potential

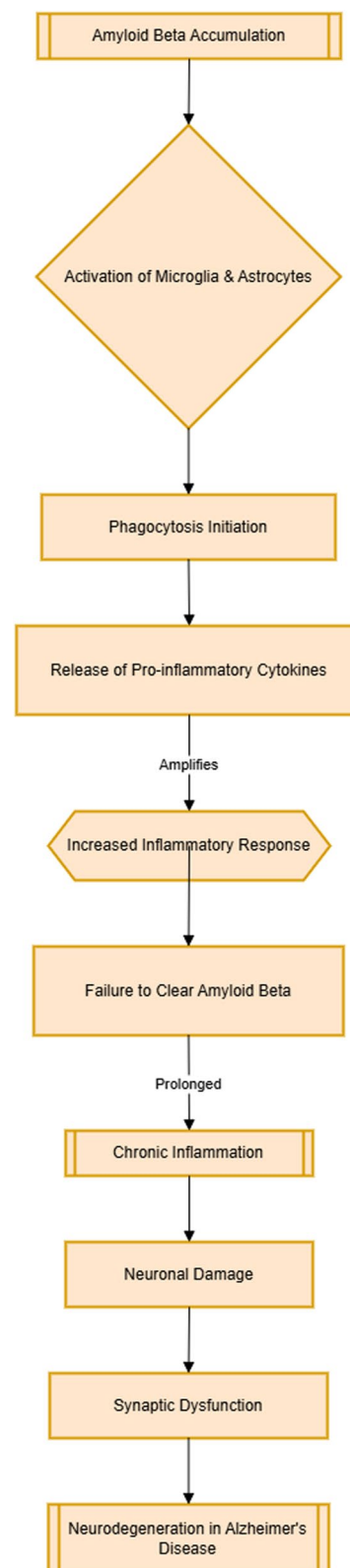


Fig. 1 Amyloid-beta accumulation leading to neurodegeneration in AD

to mitigate neuroinflammation and improve cognitive outcomes [8, 12]. Emerging oral anti-amyloid agents, such as valiltramiprosate, offer an innovative approach by interfering with the nucleation and aggregation of soluble A β oligomers independent of existing plaques. Valiltramiprosate is metabolized to tramiprosate, which prevents the coalescence of neurotoxic A β species, thus functioning via a plaque-independent pathway [13]. This mechanism contrasts with traditional immunotherapies, highlighting the expanding therapeutic landscape in AD. Additionally, newer agents, such as gantenerumab, currently under investigation, aim to refine epitope targeting and enhance immune-mediated clearance while minimizing off-target effects. The development of these agents reflects a deeper understanding of the molecular heterogeneity within AD pathology and emphasizes the need for personalized approaches [8].

Despite these advancements, significant challenges remain. The extent to which microglia selectively eliminate antibody-tagged A β species versus untagged aggregates requires further elucidation. Moreover, the long-term effects of altering immune pathways in the central nervous system and the potential for adverse immune responses necessitate careful evaluation. Optimizing the balance between efficacy and safety in immunotherapy remains a critical research focus. Alzheimer's disease remains a formidable public health challenge, yet recent therapeutic breakthroughs provide hope for altering its trajectory. Monoclonal antibodies like lecanemab, alongside innovative agents such as valiltramiprosate, exemplify the progress in targeting the underlying pathology of AD [13]. These advancements underscore the importance of continued research into adaptive immune mechanisms and the development of multi-modal treatment strategies to combat this devastating disease.

Neurodegenerative cascade of AD

The neuropathological hallmark of Alzheimer's disease (AD) encompasses amyloid plaque deposition, neurofibrillary tangle (NFT) accumulation, and cerebral amyloid angiopathy, accompanied by pronounced glial activation and significant neuronal and synaptic degeneration [14]. The amyloid accumulation hypothesis posits that a disequilibrium between protein synthesis and degradation underlies AD pathogenesis [15]. Physiologically, amyloid precursor protein (APP) is cleaved by β - and γ -secretases, generating A β peptides that may initiate amyloidogenic pathways under dysregulated conditions [16]. In AD, aberrant APP processing results in the disproportionate accumulation of A β 42 over A β 40, with the former aggregating into extracellular senile plaques. However, the mere presence

of A β plaques does not fully account for the complexity of AD pathogenesis, as many individuals with significant amyloid deposition remain cognitively unimpaired, suggesting additional pathological contributors [15].

An analytical assessment of the amyloid accumulation hypothesis reveals substantial shortcomings in its reductionist perspective AD etiology. The ACH oversimplifies the intricate interaction of genetic, vascular, and lifestyle-related risk factors by emphasizing amyloid- β (A β) accumulation [17–19]. Moreover, it neglects the physiological roles of amyloid precursor protein (APP), A β , and presenilin-1 (PS1), disregarding their evolutionary importance and complex processing routes. The reliance on preclinical models, particularly transgenic animals with familial Alzheimer's disease (FAD) mutations, is problematic due to species differences in ageing processes and brain development [3, 20–22]. This raises questions about the translational applicability of these models. Furthermore, the ACH has not definitively identified the deleterious molecular form of A β , its exact location, or precise mode of pathogenesis [4, 23–29]. The absence of a precise causal mechanism contradicts the idea and obstructs the advancement of viable treatments focused on A β reduction. Genetic evidence challenges the primary function of A β in AD etiology, as FAD mutations are absent in α - and β -secretases and crucial A β -degrading metalloproteases [6]. The restricted efficacy of anti-amyloid therapies highlights the need for a redefined comprehension of AD [7]. These critical perspectives underscore the deficiencies in the current understanding of the amyloid hypothesis. Future research should employ a more holistic approach, investigating the complex interplay among multiple factors in AD pathogenesis and considering alternate theories.

Recent advancements in AD research are broadening perspectives beyond the amyloid-centric view, placing greater emphasis on the contributions of tau pathology and neuroinflammation. The "tau propagation hypothesis" posits that the misfolding of tau proteins and their subsequent spread across synaptically connected neurons drive neurodegeneration more directly than amyloid plaques. This shift highlights the role of tau as a central mediator of neuronal loss and cognitive impairment. Parallel to this, the role of non-neuronal cells, particularly microglia, has garnered attention. Emerging evidence redefines microglia-mediated inflammation as a potential upstream driver of neurodegenerative cascades, rather than a secondary response to amyloid accumulation [30]. This paradigm recognizes microglial activation as a double-edged sword: while necessary for amyloid clearance, prolonged activation may exacerbate synaptic

dysfunction and neuronal damage, further fueling the disease.

AD pathogenesis is increasingly understood as a multifactorial process involving bioenergetic dysfunction, oxidative stress, and protein misfolding. The interplay between amyloid- β (A β) accumulation, tau pathology, oxidative stress, and neuroinflammation in AD is complex and multifaceted. A β accumulation triggers microglial activation, which leads to the release of pro-inflammatory cytokines and reactive oxygen species (ROS), which in turn exacerbate tau pathology and oxidative stress [31]. Activated microglia have been found to contribute to tau hyperphosphorylation through various pathways, including the IL-1 β /p38 α MAPK pathway [32]. The hyperphosphorylation of tau facilitates the formation of neurofibrillary tangles, a hallmark feature of AD. Moreover, oxidative stress, caused by both A β and activated microglia, can further exacerbate tau phosphorylation and aggregation [33]. Notably, tau pathology can also stimulate microglia, creating a vicious cycle of neuroinflammation and tau accumulation. Recent studies have demonstrated that the NLRP3 inflammasome, activated by aggregated tau seeds in microglia, can exacerbate tau pathology and its dissemination [34–37]. These findings highlight the intricate network of pathogenic mechanisms underlying AD etiology, emphasizing the need for multi-target therapeutic strategies to effectively treat this complex disease.

Furthermore, it has been observed that ROS-induced toxicity disrupts mitochondrial homeostasis, compounding the deleterious effects of A β 42 and tau aggregates. Moreover, immune dysregulation, characterized by sustained microglial activation and cytokine release, creates a neuroinflammatory milieu that accelerates synaptic and neuronal injury [38]. Notably, this complex interplay among cellular pathways underscores the need for integrated therapeutic strategies that address these overlapping mechanisms. Innovative treatment approaches are now targeting mitochondrial resilience, tau aggregation, and neuroinflammation to counteract the multifaceted pathology of AD.

Mitochondrial resilience has emerged as a promising therapeutic target for early intervention in AD, substantiated by recent research findings [39]. This approach is grounded in the understanding that mitochondrial dysfunction is a critical component of AD pathology, encompassing oxidative stress-related respiratory chain dysfunction, diminished mitochondrial biogenesis, and mtDNA mutations [39, 40]. The mitochondrial cascade hypothesis provides a robust framework for investigating the potential of enhancing mitochondrial resilience. This hypothesis

posits that individuals possess an inherited baseline level of mitochondrial function and resilience, which, when exceeded by decline, initiates AD-related pathological changes, including A β deposition [40]. Early mitochondrial dysfunction is associated with increased A β production and aggregation, highlighting the importance of targeting mitochondrial resilience in the early stages of the disease process. Various strategies aimed at enhancing mitochondrial function have shown promise in preclinical investigations, including the development of mitochondria-targeted antioxidants, enhancement of mitochondrial biogenesis via PGC-1 α activation, and inhibition of excessive mitochondrial fission [41–45]. Gene therapy strategies targeting mtDNA mutations also demonstrate potential in enhancing the proportion of wild-type mtDNA relative to mutant mtDNA [41–45]. Emphasizing mitochondrial resilience is crucial for advancing preventive treatment approaches in AD. Early intervention aimed at enhancing mitochondrial function may provide a more effective strategy for disease prevention and management.

Diagnostic markers for Alzheimer's disease

Key diagnostic instruments for AD include imaging techniques such as magnetic resonance imaging (MRI), 18 fluorine-fluorodeoxyglucose (18F) positron emission tomography, amyloid-beta positron emission tomography, Tau protein positron emission tomography (18F-flortaucipir PET) as well as CSF biomarkers (A β 42, Tau) [46, 47].

Recent advances in neuroimaging techniques, including computerized tomography (CT), magnetic resonance imaging (MRI), and positron emission tomography (PET), have significantly enhanced our understanding of AD pathology [48, 49]. These imaging modalities provide valuable insights into brain structure and function, enabling the detection of neuronal loss, atrophy patterns, and metabolic impairments characteristic of AD. MRI has demonstrated considerable promise in distinguishing AD-related atrophy from normal aging and differentiating AD from other forms of dementia [49]. This imaging modality offers a non-invasive means of assessing brain atrophy patterns, which can aid in the diagnosis and monitoring of AD progression. PET imaging, particularly when employed with specific tracers such as Pittsburgh compound B (PIB), has shown efficacy in detecting amyloid deposition patterns typical of AD [50, 51]. This imaging technique provides a valuable tool for visualizing amyloid plaques, a hallmark of AD pathology. Despite the potential of these neuroimaging modalities, several limitations exist. These techniques are often expensive, require specialized equipment and expertise, and may not always provide satisfactory sensitivity and specificity

for AD diagnosis [52]. Further research is necessary to address these limitations and optimize the utility of neuroimaging techniques in AD diagnosis.

The accuracy and precision of amyloid-PET investigations for the ascertainment of AD are approximately 90% [53, 54]. Tau-PET has been demonstrated to be more accurate than MRI and effective in distinguishing AD from other neurodegenerative conditions [55]. Preclinical AD can be detected in asymptomatic patients through amyloid-PET, often long before the emergence of noticeable clinical manifestations [56].

CSF biomarkers, including A β 42, P-Tau, and T-Tau, have been shown to be highly diagnostic for Alzheimer's disease (AD), with an accuracy of over 90% when used in conjunction [57]. However, both positron emission tomography (PET) and CSF diagnoses are associated with significant challenges. PET is a costly procedure, whereas CSF sampling, although more economical, is invasive and time-consuming [53, 58]. These limitations highlight the need for less invasive, cost-effective approaches for the initial identification of AD and for differentiating it from other neurocognitive impairment diseases [59, 60]. Furthermore, the diagnostic utility of CSF markers is restricted by their limited specificity. For instance, elevated levels of total tau (t-tau) in CSF, although sensitive for detecting AD, are not specific to the disease and can also be observed in individuals with vascular dementia and frontotemporal dementia [61]. Therefore, the development of novel, non-invasive, and cost-effective biomarkers for AD is essential for improving diagnostic accuracy and facilitating early intervention.

Blood-based indicators, particularly plasma Tau, are advantageous as they are more suitable for primary healthcare environments because to simpler sample procedures [62]. That said, heterogeneity in plasma indicators continues to be a concern. The hydrophobic attributes of A β peptides present considerable analytical difficulties, especially regarding plasma-based assessments. A β 's propensity to associate with plasma proteins may result in epitope masking and various analytical interferences, hindering the precise determination of A β concentrations in plasma [63]. The plasma A β measurements are susceptible to confounding influences from systemic factors extrinsic to cerebral A β production, thereby potentially introducing discrepancies between plasma A β levels and actual brain A β pathology, and compromising diagnostic accuracy [63, 64]. Notwithstanding the aforementioned limitations, recent studies have yielded promising results, underscoring the potential of blood-based biomarkers for Alzheimer's disease (AD) diagnosis. For instance, a notable investigation analyzing a panel of 18 plasma signaling

and inflammatory proteins demonstrated encouraging results in distinguishing AD patients and predicting AD onset in individuals with mild cognitive impairment [65]. However, these findings necessitate further validation through rigorous research endeavors to establish more reliable and robust blood-based biomarkers for AD. MicroRNAs (miRNAs) in plasma, serum, and cerebrospinal fluid are increasingly recognized as possible metrics for AD and other conditions [66–69]. The inflammatory response in Alzheimer's disease (AD) is closely intertwined with alterations in microRNA (miRNA) expression profiles. Elucidating the specific miRNAs involved in the regulation of pro-inflammatory cytokine and proteolytic enzyme production is crucial for understanding their role in AD pathogenesis. Investigating the enzymatic machinery implicated in this process is essential for deciphering their functional contributions to disease progression.

Recent studies have demonstrated that the overexpression of miR-146a in microglial cells can mitigate cognitive impairments, reduce neuroinflammatory responses, decrease amyloid- β (A β) levels, and prevent neuronal demise. This therapeutic effect is mediated by promoting a phenotypic shift from the pro-inflammatory M1 microglial subtype to an anti-inflammatory phenotype, thereby modulating the neuroinflammatory milieu and alleviating AD-related pathology [70, 71]. miRNAs in plasma have emerged as promising biomarkers for the diagnosis of AD, offering a non-invasive and potentially more accessible method for diagnosis [72]. Studies have shown that miRNAs are aberrantly expressed in AD, exhibiting significant correlations with hallmark pathological characteristics, including amyloid- β deposition, tau hyperphosphorylation, and neuroinflammatory responses. Li, 2024 #68} A growing body of evidence supports the utility of plasma microRNAs (miRNAs) as diagnostic biomarkers for AD. A comprehensive meta-analysis of blood exosome studies revealed a significant downregulation of miR-212 and miR-132 in AD patients compared to healthy controls. Conversely, elevated levels of exosomal miR-135a, miR-193b, and miR-384 were detected in the serum of AD patients. Notably, miR-132 has emerged as a promising biomarker, demonstrating a robust capacity to differentiate AD patients from controls in blood-based assays, with high sensitivity and specificity [73–75]. This mechanistic interplay between miRNA dysregulation and AD pathology highlights the potential of miRNAs as sensitive and informative biomarkers for disease diagnosis and monitoring. However, further research is needed to standardize miRNA measurement techniques and investigate the specificity of individual miRNAs for AD versus other neurodegenerative diseases.

MiRNAs play crucial roles in regulating gene expression related to AD pathology, including amyloid- β production, tau phosphorylation, and neuroinflammation.

The application of miRNAs as biomarkers in AD research faces several significant limitations that warrant careful consideration. A primary concern is the issue of reproducibility across studies. Despite using similar sample sources and subject demographics, researchers have encountered considerable inconsistencies in miRNA biomarker results, hindering the establishment of reliable miRNA biomarkers for AD [76–78]. Methodological heterogeneity poses a significant obstacle to advancing miRNA biomarker research in AD. The lack of standardized operating procedures for sample acquisition, miRNA extraction, detection methodologies, and data interpretation contributes to inconsistencies in results. Variables such as cellular contamination, hemolytic events, and disparities in quantification approaches can profoundly influence outcomes, thereby hindering inter-study comparisons and highlighting the need for harmonized protocols to ensure result reproducibility and validity [79]. Endogenous variability complicates miRNA [80] biomarker research, as miRNAs from different biological sources can confound results. Platelet-derived miRNAs, for instance, can affect serum and plasma concentrations, emphasizing the need for careful sample selection and processing [80].

Superior clarity eye scans, including optical coherence tomography (OCT), demonstrate potential for finding the initial signs of AD. Research involving experimental settings of AD and human subjects has identified degenerative retinal alterations that are associated with early neurodegeneration [81–91]. OCT facilitates high-resolution, non-invasive visualization of retinal structures, with particular emphasis on the retinal nerve fiber layer (RNFL) and ganglion cell layer (GCL). A burgeoning body of evidence from multiple studies has consistently demonstrated significant RNFL thinning in Alzheimer's disease (AD) patients relative to healthy controls, with pronounced effects observed in the superior and inferior quadrants. Notably, this structural alteration has been correlated with cognitive decline in select studies, suggesting that retinal degeneration may serve as a surrogate marker for neurodegenerative processes in the brain, warranting further investigation into the relationship between retinal pathology and AD [92–94]. Research has yielded additional insights into the retinal manifestations of Alzheimer's disease (AD), with studies demonstrating macular thinning and reduced macular volume in AD patients, which may be correlated with disease progression. Moreover, these retinal alterations have also been observed in patients with mild cognitive impairment (MCI), albeit with less severity

compared to AD patients. The detection of retinal thinning in MCI patients suggests that optical coherence tomography (OCT) may possess utility in identifying early stages of cognitive decline, thereby potentially enabling earlier intervention and diagnosis [95].

The current evidence base for the use of optical coherence tomography (OCT) in Alzheimer's disease (AD) diagnosis is limited by several factors. Firstly, many studies have been conducted with relatively small sample sizes, which may not be representative of the broader AD population. Furthermore, there is a degree of inconsistency in findings across different patient populations, with some studies reporting significant correlations between OCT measurements and cognitive scores, while others have failed to replicate these associations [96–99]. Additionally, the specificity of OCT for AD diagnosis is potentially compromised by the observation of similar retinal changes in other neurodegenerative conditions, such as Parkinson's disease [100]. These limitations underscore the need for further research to standardize OCT protocols, increase sample sizes, and clarify the diagnostic specificity of OCT in AD.

A β constitutes a critical element of amyloid aggregates in the cerebral tissue of individuals with AD, where A β 42 exhibits a significant association with pathogenicity and A β 40 is involved in the formation of plaque cores [101]. Plasma A β has potential as a cost-effective diagnostic tool; however, its clinical application is constrained by variable results and lower plasma concentrations relative to cerebrospinal fluid. Hypersensitive techniques such as SIMOA and IP-MS enhance detection accuracy; however, plasma A β demonstrates optimal effectiveness when integrated with additional diagnostic methods [102, 103].

Microtubule-associated tau assumes a pivotal function in maintaining neuronal structural integrity; however, in AD, it forms aggregates known as neurofibrillary tangles. Elevated levels of P-Tau, particularly P-Tau181 and P-Tau217, are indicative of AD and can be identified in CNS and peripheral fluid biomarker [104–106]. Plasma Tau concentrations are notably lower than those in cerebrospinal fluid (CSF); however, advancements in detection methods such as SIMOA and Meso-Scale have enhanced measurement capabilities [106–108]. Plasma P-Tau exhibits a strong correlation with cognitive decline, brain atrophy, and hypometabolism [109, 110].

Neurogranin, a calmodulin-binding protein, serves as a marker for synaptic loss and is found to be amplified in cerebrospinal fluid of AD patients [111–116]. This distinctive neuropathological feature is characteristic of Alzheimer's disease and correlates with a deterioration in cognitive function and brain atrophy [117, 118]. Neurofilament light (NFL) serves as a marker for axonal

damage and demonstrates a strong correlation with visuospatial deterioration and the severity of AD [119–121]. Plasma NFL levels are valuable for monitoring neurodegeneration; however, they lack specificity for AD, as their increased expression is noted across diverse brain disorders [122–125].

Chronic neuroinflammation is a significant element to consider progression of AD. Genes such as ABCA7, CD33, CR1, EPHA1, MS4, and TREM2 participate in the inflammatory response [126]. Increased concentrations of pro-inflammatory molecules correlate with Alzheimer's disease; however, additional studies are necessary to determine their diagnostic significance.

The incorporation of multiple biomarkers, including imaging and fluid-based measures, has the potential to significantly enhance diagnostic precision in Alzheimer's disease (AD). This multi-modal approach is aligned with the Amyloid Tau Neurodegeneration (A,T,N) Research Framework, which advocates for the utilization of a combination of biomarkers to facilitate AD diagnosis and characterization [127]. The framework integrates three primary biomarker categories: amyloid pathology (A), assessed via amyloid positron emission tomography (PET) or cerebrospinal fluid (CSF) amyloid- β (A β) protein; tau pathology (T), evaluated using tau PET or CSF phosphorylated tau (p-tau); and neurodegeneration (N), measured through magnetic resonance imaging (MRI) atrophy, CSF total tau (t-tau), or fluorodeoxyglucose (FDG) PET [128, 129].

By combining these diverse biomarker types, researchers and clinicians can gain a more comprehensive understanding of the complex biological processes underlying AD. For instance, the concurrent use of amyloid-PET and CSF A β measurements can provide complementary insights into amyloid pathology. Furthermore, the integration of fluid biomarkers, such as plasma phosphorylated tau (P-Tau), with imaging biomarkers may offer a less invasive and more accessible initial screening method, which can subsequently be confirmed using more specific imaging techniques [130].

This multi-modal approach can facilitate more accurate identification of AD pathology, disease staging, and differentiation from other neurodegenerative disorders. While individual biomarkers may have limitations, their combination can increase diagnostic confidence and provide a more complete picture of the disease state. This integrated approach is particularly valuable in the early stages of AD, where symptoms may be subtle or absent, but biological changes are already underway.

A promising advancement in the therapy of Alzheimer's disease: a scrutiny of the parameters of lecanemab

Lecanemab, an anti-A β monoclonal antibody, exhibits a targeted strategy by preferentially interacting with soluble A β oligomers and fibrillar aggregates. This novel class of monoclonal antibodies selectively neutralizes large-sized soluble A β protofibrils and fibrillar aggregates, which are associated with enhanced neurotoxicity, synaptic dysfunction, and electrophysiological disturbances in AD [132]. Distinct from existing anti-amyloid therapies, lecanemab directly mitigates neuronal hyperexcitability and restores neural oscillations, thereby alleviating memory impairments [132]. Notably, the APP Arctic variant, which enhances production of toxic A β protofibrils linked to early-stage AD pathogenesis [132], underscores the cytotoxic consequences of A β polymerization and informs therapeutic strategies targeting these species [131]. Lecanemab is uniquely engineered to bind both soluble protofibrils (exceeding monomer size) and insoluble fibrils in amyloid plaques, reducing neurotoxic burden across A β forms [132–138].

Compared to other monoclonal antibodies, lecanemab demonstrates superior affinity for soluble protofibrils—approximately 100-fold greater than aducanumab for small protofibrils and 25-fold greater for larger ones, with minimal monomer binding [139]. Aducanumab shows limited interaction with small oligomers, while gantenerumab binds oligomers and monomers more strongly but with lower selectivity [140]. Donanemab, conversely, targets insoluble plaques exclusively [141].

In clinical trials, lecanemab significantly reduces A β burden in early AD versus placebo [142], with a phase 2 study reporting cerebral amyloid-related imaging changes (CARICs) in 21.5% of patients (for example, vasogenic edema, microbleeds) [143]. ARIA-E incidence remained below 10%, with symptomatic edema in <3% of participants receiving 10 mg/kg biweekly infusions [144]. Despite discontinuation of high-dose regimens for ApoE4 carriers (70% of the cohort), lecanemab's ARIA rates are lower than other antibodies, with asymptomatic ARIA-E typically mild-to-moderate and occurring early [145].

The phase 3 CLARITY-AD trial demonstrated a 27% slowing of clinical decline over 18 months (CDR-SB) versus placebo, though effect sizes were modest. Age stratification revealed disparities: patients aged 50–64 saw 6% slowing versus 40% in those >75 years, highlighting questions about intervention timing and early-onset AD management [146].

Lecanemab's safety profile is favorable relative to other amyloid-targeting antibodies, with lower ARIA incidence

(ARIA-E/ARIA-H), broadening eligibility for patients with comorbidities or anticoagulant use [143, 146, 147]. However, APOE $\epsilon 4$ homozygosity elevates ARIA risk, necessitating genetic testing and individualized monitoring [148, 149]. APOE $\epsilon 4$ carriers' higher amyloid burden may enhance therapeutic response but requires balanced risk mitigation [150].

Clinical implementation demands early AD diagnosis with confirmed amyloid pathology, multidisciplinary care, and realistic expectations (modest cognitive/functional delay, not a cure) [151]. Lecanemab's specificity for soluble protofibrils—key neurotoxic agents—likely underpins its efficacy and reduced ARIA risk versus fibril-targeting antibodies [152]. Long-term real-world data and predictive models integrating genetic/biomarker profiles are critical for optimizing personalized outcomes [147].

Lecanemab clinical trials: insights and outcomes

Lecanemab has undergone extensive clinical trials to critically assess its therapeutic efficacy, optimal dosing, and safety evaluation for early AD. A phase 2, exploratory study examined the efficacy and safety of lecanemab in individuals with mild Alzheimer's pathology, characterized by early signs of memory decline and neurodegenerative changes in approximately 900 individuals with mild Alzheimer's pathology, characterized by prodromal cognitive impairment and mild neurodegenerative dementia symptoms [145, 153]. The trial employed a Bayesian framework with adaptive randomization guided by responses to evaluate three dosing regimens against a comparator. The aim was to identify the minimum effective dose (ED90) that achieves 90% of the maximum therapeutic effect. Following the main study, an additional phase was conducted to further assess outcomes [145].

The major objective of the phase 2 the investigation was to measure the advancement of functional decline over a 12-month duration, employing a composite scoring system specifically developed for evaluating Alzheimer's-related symptoms. The study targeted an 80% probability of attaining a reduction in decline above 25% relative to the control group [145]. Secondary outcomes included PET imaging for amyloid reduction, cognitive assessments (ADCOMS, CDR-SB, ADAS-Cog), together with biomarkers in neurological fluid (CSF) and changes in hippocampal volume, aiming to capture a multidimensional view of lecanemab's impact on AD progression [145]. The biweekly administration of lecanemab at a designated dosage (10 mg/kg) demonstrated a substantial probability of attaining the primary objective within one year. By 18 months, this dose further demonstrated promising

efficacy, reducing ADCOMS scores by 27–30%, ADAS-Cog by 47–56%, and CDR-SB by 26–33%, with notable reductions in amyloid burden as visualized on PET [145]. Although the study's primary Bayesian endpoint at 12 months was unmet, the 18-month analysis demonstrated substantial dose-responsive benefits across both clinical and biomarker measures. Based on these findings, the optimal 10 mg/kg biweekly dose without titration was selected for the phase 3 *Clarity AD* trial and open-label extension, aiming to further validate lecanemab's amyloid-targeting and clinical efficacy in early AD [145]. This nuanced dose–response relationship, evidenced by a progressive decline in clinical scores and amyloid burden beyond the primary endpoint period, reflects lecanemab's potential as a disease-modifying therapy that requires sustained dosing for maximal benefit. These findings underscore the importance of longitudinal analysis in AD treatment trials to capture evolving drug effects and refine dosing strategies for optimal patient outcomes in AD management. Lecanemab demonstrated good tolerability, with prevalent adverse events comprising amyloid-related cerebral abnormalities and drug administration complications. ARIA with edema (ARIA-E) occurred in less than 10% of patients receiving higher doses and in 14.3% of ApoE4 carriers, typically manifesting within the initial three months and predominantly classified as mild to moderate [145]. The decreased incidence of ARIA may be associated with lecanemab's diminished affinity for A β fibrils [141, 154–157].

The phase 3 *Clarity AD* trial confirmed these results, demonstrating that lecanemab achieved all primary and secondary endpoints. In a study involving 1,795 participants, lecanemab demonstrated a 27% reduction in CDR-SB decline at 18 months relative to placebo, with observable benefits emerging as early as 6 months [146]. The reduction in brain amyloid burden was significant, exhibiting a mean difference of -70 centiloids ($P < 0.001$). Lecanemab demonstrated a 26% reduction in cerebral performance changes as evaluated by ADAS-Cog14 and a 37% reduction in functional loss according to ADL measures [146]. In the lecanemab group, the occurrence of edema associated with amyloid pathology was markedly higher in the lecanemab group compared to the placebo arm [146]. The phase 3 trial indicated that lecanemab significantly decreased cognitive impairment, functional decline, and amyloid accumulation in early Alzheimer's disease [146]. The clinical significance of lecanemab's therapeutic benefits in early Alzheimer's disease (AD) must be carefully evaluated in the context of its limitations. While the phase 3 trial outcomes suggest that lecanemab may exert disease-modifying effects,

Table 1 Comprehensive inclusion and exclusion parameters utilized in lecanemab trials, emphasizing patient selection criteria to ensure participant safety, homogeneity, and alignment with study objectives

Eligibility parameters utilized in the study	Criteria for patient selection in lecanemab trials	Significance
MCI or mild AD	Mild neurocognitive disorder, as defined by established diagnostic criteria and assessed through severity measures, encompasses mild cognitive dysfunction due to AD [158, 159].	Targets early AD stages, enabling therapeutic interventions to slow disease progression
Episodic memory deficit	Mild neurocognitive disorder, as defined by established diagnostic criteria and assessed through severity measures encompasses mild cognitive dysfunction due to AD [158, 159].	Establishes robust cognitive benchmarks for inclusion, ensuring homogeneity of participant profiles
Amyloid biomarkers	Positive amyloid-PET or cerebrospinal fluid (CSF) results confirming AD pathology	Integrates biomarker-based precision, aligning with evolving trends in AD diagnostic practices
Age	Participants aged 50–90 years, with exceptions at investigator discretion	Ensures a broad yet clinically relevant sample, balancing generalizability and specificity
MMSE score	Scores between 22–30 at screening and baseline, indicating early-stage AD	Correlates cognitive function with disease stage, refining participant stratification
BMI	Body mass index (BMI) between 17 and 35, with clinical judgment applied to extremes	Incorporates physical health parameters while accommodating outliers for broader applicability
Stable AD medications	Stable use of acetylcholinesterase inhibitors or memantine for ≥ 12 weeks before baseline	Maintains pharmacological stability, minimizing variability in baseline cognitive measures
Study partner	Requires an identified partner for treatment adherence and study participation	Acknowledges the role of caregivers in supporting long-term clinical and research goals
Informed consent	Obtaining documented consent with knowledge from individuals and study collaborators, assuring their understanding of the associated risks and advantages	Reinforces ethical principles, addressing informed decision-making in AD research participation
Exclusion criteria		
Non-AD cognitive issues	Neurological or psychiatric conditions unrelated to AD that affect cognition	Focuses on AD-specific pathology, ensuring validity of trial outcomes by eliminating confounders
Brain abnormalities	> 4 microhemorrhages, macrohemorrhages > 10 mm, or significant MRI findings inconsistent with AD	Enhances participant safety by excluding individuals with brain abnormalities unrelated to AD
Recent stroke/seizures	A previous record of cerebrovascular incidents, temporary neurological abnormalities, or seizure episodes within the preceding year	Mitigates risks of cerebrovascular events during treatment, safeguarding participant outcomes
Psychiatric conditions	Severe psychiatric disorders (for example, psychosis, uncontrolled depression)	Eliminates variables that may interfere with treatment adherence and accurate cognitive assessments
Immunological diseases	Uncontrolled autoimmune conditions or use of immunosuppressive therapies	Addresses potential risks of immunological complications related to the therapeutic mechanism
Bleeding disorders	Uncontrolled coagulopathies or unstable anticoagulant therapy	Ensures safety in participants at risk for hemorrhagic events
Systemic illnesses	Unstable cardiac, renal, gastrointestinal, or respiratory conditions	Excludes conditions that could confound results or exacerbate treatment-related risks

the modest effect sizes observed in cognitive and functional outcomes necessitate a nuanced interpretation of the findings.

The trial’s limitations, including the exclusion of individuals with comorbidities and the restricted generalizability of the results to real-world populations, highlight the need for further research to elucidate the therapeutic profile of lecanemab. Moreover, the long-term safety and efficacy of lecanemab remain to be fully characterized, particularly in diverse patient populations and over extended treatment periods.

A critical examination of the trial’s methodology and results is essential to inform the design of future studies and to optimize the clinical use of lecanemab in early AD. Furthermore, ongoing research should prioritize the development of personalized treatment strategies, incorporating biomarkers and other prognostic indicators to tailor treatment approaches to individual patient needs.

Table 1 shows the eligibility criteria for lecanemab clinical trials in AD.

Imaging protocol guidance for lecanemab administration

To determine eligibility for lecanemab treatment, unenhanced magnetic resonance imaging is recommended, employing T1 fluid-attenuated inversion recovery (FLAIR), T2*-weighted gradient recalled echo (GRE) or comparable protocols [160]. A prior MRI (within 12 months) with these sequences can be employed to assess candidacy for lecanemab therapy, assuming abnormal amyloid status is confirmed and other specified criteria are fulfilled.

Imaging evaluations are an essential diagnostic tool for identifying potential contraindications for lecanemab treatment. Patients with substantial cerebrovascular pathology, such as large cerebral hemorrhages or multiple small hemorrhages suggesting vascular fragility, should be excluded from treatment. Furthermore, individuals with superficial siderosis, characterized by iron deposition in the brain, or those displaying brain edema due to vascular permeability, are not suitable candidates. Extensive white matter disease, multiple small vessel strokes, and large territorial strokes impacting essential brain regions are additional contraindications that necessitate exclusion. Cerebrovascular conditions may increase the risk of adverse events or undermine the safety and efficacy of lecanemab treatment [160]. Additionally, documentation of cerebral pathology, including traumatic injury, degenerative changes, or vascular anomalies also warrants exclusion. Furthermore, patients with MRI evidence of underlying cerebral (amyloid angiopathy) or conditions increasing the risk of amyloid-related

imaging abnormalities (ARIA) should not receive lecanemab treatment [160]. MRI surveillance of amyloid-related imaging abnormalities (ARIA), encompassing edema (ARIA-E) and hemorrhage (ARIA-H), critically informs clinical decision-making during anti-amyloid antibody therapy. For lecanemab, ARIA-E incidence reached 9.9% at the 10 mg/kg biweekly dose, with severity on MRI guiding dose modifications or discontinuation [145]. Asymptomatic, mild ARIA-E may permit continued treatment under enhanced monitoring, while moderate-to-severe or symptomatic cases typically necessitate temporary suspension [145, 161, 162]. Treatment resumption requires radiographic improvement, assessed via 4- to 6-week follow-up MRI [162]. Notably, ApoE4 carriers exhibit elevated ARIA-E risk (14.3% incidence with lecanemab), warranting genotype-informed dosing and monitoring protocols [144]. This genetic stratification highlights the need to balance amyloid-targeted efficacy against ARIA risk, particularly given lecanemab's demonstrated clinical benefits in early Alzheimer's disease [144, 145, 163]. These stringent MRI-based exclusion criteria aim to minimize potential threats of adverse events associated with this monoclonal antibody therapy. Table 2 shows clinical and MRI-based diagnostic criteria for probable CAA-ri, outlining key clinical symptoms, imaging characteristics, and exclusionary conditions [164].

Accommodating patients unable to undergo MRI requires a multifaceted approach informed by patient-specific needs and clinical objectives. For individuals with claustrophobia or cognitive impairments, such as

Table 2 Criteria for probable cerebral amyloid angiopathy-related inflammation (CAA-ri)

Criterion [164]	Details [164]
Age	Patient is aged 40 years or older (or has reached an age threshold of 40 years or more)
Clinical presentation	The existence of at least one of the subsequent symptoms, not directly associated with recent cerebral hemorrhage: Headache Altered level of responsiveness Alterations in conduct Localized neurological impairments Convulsive episodes
MRI findings	Imaging shows single or multiple regions of elevated white matter signal that: Exhibit asymmetry Extend into the subcortical white matter Are not attributable to prior intracerebral hemorrhage
Hemorrhagic lesions	Presence of at least one cortico-subcortical hemorrhagic lesion, such as: Large cerebral hemorrhagic lesions Small cerebral hemorrhagic lesions Superficial cortical iron deposition
Exclusion of other causes	No evidence of neoplastic, infectious, or alternative etiologies to account for the findings

those with Alzheimer's disease (AD), advanced imaging centers equipped with open MRI systems offer a viable alternative [165]. These systems reduce the psychological discomfort associated with traditional closed-bore scanners, potentially improving compliance and scan completion rates. In cases where anxiety or agitation persists despite environmental modifications, establishing rapport with AD patients within the brief timeframe of imaging examinations (typically 15–30 min) poses significant challenges for radiographers. Integrating caregivers who share pre-existing relationships with patients can mitigate these difficulties. Empirical evidence indicates that active caregiver involvement during imaging procedures reduces agitation, minimizes disruptive behaviors, and enhances patient cooperation, thereby improving procedural efficiency and outcomes [165–170].

When MRI remains impractical, alternative imaging modalities such as positron emission tomography (PET) provide diagnostic utility. PET scans, which are typically shorter in duration and less physically restrictive, enable evaluation of brain metabolism and amyloid plaque deposition—key biomarkers in AD pathology [171]. This modality may be particularly advantageous for patients with severe movement intolerance or those requiring amyloid-specific diagnostic information. Sedation for claustrophobic Alzheimer's disease (AD) patients requiring MRI necessitates careful risk–benefit analysis. Preclinical studies suggest anesthetics may accelerate beta-amyloid aggregation and tau phosphorylation, potentially exacerbating AD pathology [2, 3, 172–175]. However, human studies remain inconclusive, with some indicating no significant association between anesthesia exposure and AD progression [176], and others suggesting potential protective effects linked to perioperative care [177]. Clinical protocols should prioritize minimal effective sedation dosages, considering drug interactions with AD medications and patient comorbidities. Current evidence supports judicious use of sedation when non-pharmacological alternatives fail, emphasizing patient-specific assessment and post-procedure monitoring. Further research is critical to clarify long-term neurological impacts and optimize sedation strategies for this vulnerable population.

Non-pharmacological interventions also play a critical role. Sensory adaptations, including environmental adjustments to enhance comfort (for example, personalized music, familiar objects, or calming visuals), have demonstrated efficacy in reducing patient distress and improving cooperation during imaging procedures. Such strategies, supported by evidence from pediatric and AD care studies, align with a patient-centered care

model that prioritizes psychological well-being alongside diagnostic accuracy [166–170, 178–182].

Ultimately, the selection of imaging strategies must balance clinical necessity with patient safety and comfort. A hierarchical approach—prioritizing non-invasive adaptations, followed by open MRI systems, sedation, and alternative modalities—ensures flexibility while addressing individual limitations. This paradigm reflects evolving trends in precision medicine, emphasizing adaptive, evidence-based protocols to optimize outcomes for vulnerable patient populations.

Patients considered for lecanemab therapy must undergo a screening MRI within 12 months of treatment initiation to assess the potential safety of therapy. This initial MRI serves as a baseline to evaluate potential contraindications and monitor for amyloid-related imaging abnormalities (ARIA) during treatment. MRI is mandatory for monitoring ARIA, and certain conditions render patients ineligible for lecanemab treatment [160, 183]. Specifically, patients with claustrophobia that precludes MRI, pacemaker or defibrillator implantation, or metal implants incompatible with MRI are excluded. Computerized tomography (CT) scans are insufficient for detecting pathology that may render a patient unsuitable for lecanemab therapy or for monitoring ARIA during treatment. As a result, CT scans are not a substitute for MRI in this context, and MRI remains the gold standard for evaluating safety and efficacy [183].

Dosage and administration

The dosing regimen for this monoclonal antibody consists of biweekly IV infusions, non-titrated dosing paradigm [183]. The recommended dose is 10 mg/kg body weight, weight-adjusted. Concentrations of 500 mg/5 mL and 200 mg/2 mL (100 mg/mL) are reconstituted in 250 mL 0.9% saline and infused intravenously through a 0.2- μ m filter with minimal protein binding [142, 183]. To adverse events associated with intravenous administration, a 3-h observation period is mandatory after administering the initial study medication, thereafter accompanied by an inquiry via phone to assess for delayed responses. For subsequent treatments, observation periods decrease to 90 min for doses 2–3 and 15 min for subsequent doses, unless adverse events necessitate extended monitoring. In case of any dose-missed scenarios, proceed with the next dose administration at the earliest practicable time point, maintaining the biweekly schedule [142, 184, 185]. Table 3 outlines administration procedures for investigational intravenous therapy, including dosage specifications, preparation with diluents, infusion timing, monitoring protocols for adverse reactions, and instructions for managing missed doses.

Table 3 Comprehensive administration protocol for lecanemab

Parameter [183, 185]	Details [142, 183]
Administration route	Intravenous (IV)
Dosage	10 mg/kg body weight, weight-adjusted
Vial concentrations	500 mg/5 mL (100 mg/mL) or 200 mg/2 mL (100 mg/mL)
Diluent	250 mL of 0.9% sodium chloride injection
Infusion time	Approximately 1 h
Infusion filter	Low-protein binding 0.2 micron filter
Frequency of administration	Every other week (biweekly)
Observation after infusion	First infusion: 3 h Second and third infusions: 2 h Subsequent infusions: 30 min if no reactions
Delayed reaction Monitoring	Telephone call after first infusion to monitor delayed reactions
Missed dose Instructions	Administer as soon as possible; maintain biweekly schedule

Adverse effects

Research on the tolerability of lecanemab has revealed infusion-related problems and brain abnormalities associated with amyloid buildup as the principal safety issues requiring meticulous monitoring and management [163, 183]. Patient monitoring post-treatment focuses on early detection and management of ARIA to reduce the risk of progression or recurrence [186]. Figure. 2 outlines the outlining imaging protocols, clinical assessment, and treatment modifications for amyloid-related imaging abnormalities (ARIA) during lecanemab therapy, emphasizing safety and continuity of care. Enhanced vigilance is particularly critical for APOE4 carriers, especially homozygotes, who are at elevated risk [145, 187]. After the first 12 months of therapy, the decision to perform further MRIs should be symptom-driven and guided by prior imaging results [183, 188]. Symptomatic ARIA manifests with clinical symptoms including headache, disorientation, visual abnormalities, dizziness, nausea, gait instability, and, in extreme instances, convulsions, protracted seizures, brain dysfunction, and localized neurological deficits [189]. Cerebral imaging abnormalities related to amyloid clearance are a notable side effect that has been observed in patients treated with beta-amyloid-targeting monoclonal antibodies. Amyloid-related imaging abnormalities can manifest in two distinct forms: edema-related abnormalities and hemorrhage-related abnormalities. In severe cases, cerebral macrohemorrhage can occur, leading to lasting cognitive and motor impairments. The administration of anticoagulant therapy to patients with pre-existing cerebral microvascular lesions or conditions characterized by amyloid-driven vascular inflammation, such as cerebral amyloid angiopathy-related vasculitis or amyloid β -associated vasculitis,

is associated with an elevated risk of catastrophic cerebral hemorrhage [190, 191].

Although ARIA-related events resulting in severe morbidity or mortality are uncommon with lecanemab, three fatal cases were reported during the CLARITY-AD open-label extension phase. In one instance, an elderly APOE4 non-carrier with cardiovascular disease on anticoagulation developed a macrohemorrhage. Another case involved an APOE4 homozygote who experienced severe CAA with vasculitis, multifocal hemorrhages, and a large vessel occlusion after receiving tissue plasminogen activator (tPA). The third death occurred in an APOE4 homozygote with severe ARIA-E and ARIA-H, presenting a clinical syndrome resembling CAA-ri [183, 192]. Given these risks, it is recommended that lecanemab not be used in patients receiving anticoagulants. If anticoagulation becomes necessary during treatment, lecanemab infusions should be discontinued and may be resumed only when anticoagulation is no longer required [193]. Additionally, spontaneous ARIA-like events are known to occur in patients with CAA or ABRA, conditions commonly associated with cerebral microbleeds, a known risk factor for ARIA [160, 194–196]. Patients with underlying CAA-ri are particularly vulnerable to ARIA [197]. Providers should recognize and manage rare but severe ARIA cases, resembling CAA-ri syndrome [194], characterized by extensive edema, tissue swelling, and microhemorrhages. APOE4 carriers, especially homozygotes, are at highest risk. Severe manifestations include seizures, status epilepticus, encephalopathy, stupor, and lateralizing signs. Prompt MRI scanning and hospitalization or critical care unit admission are crucial. Treatment involves high-dose glucocorticoids (for example, methylprednisolone) and seizure monitoring. Severe ARIA may be reversible

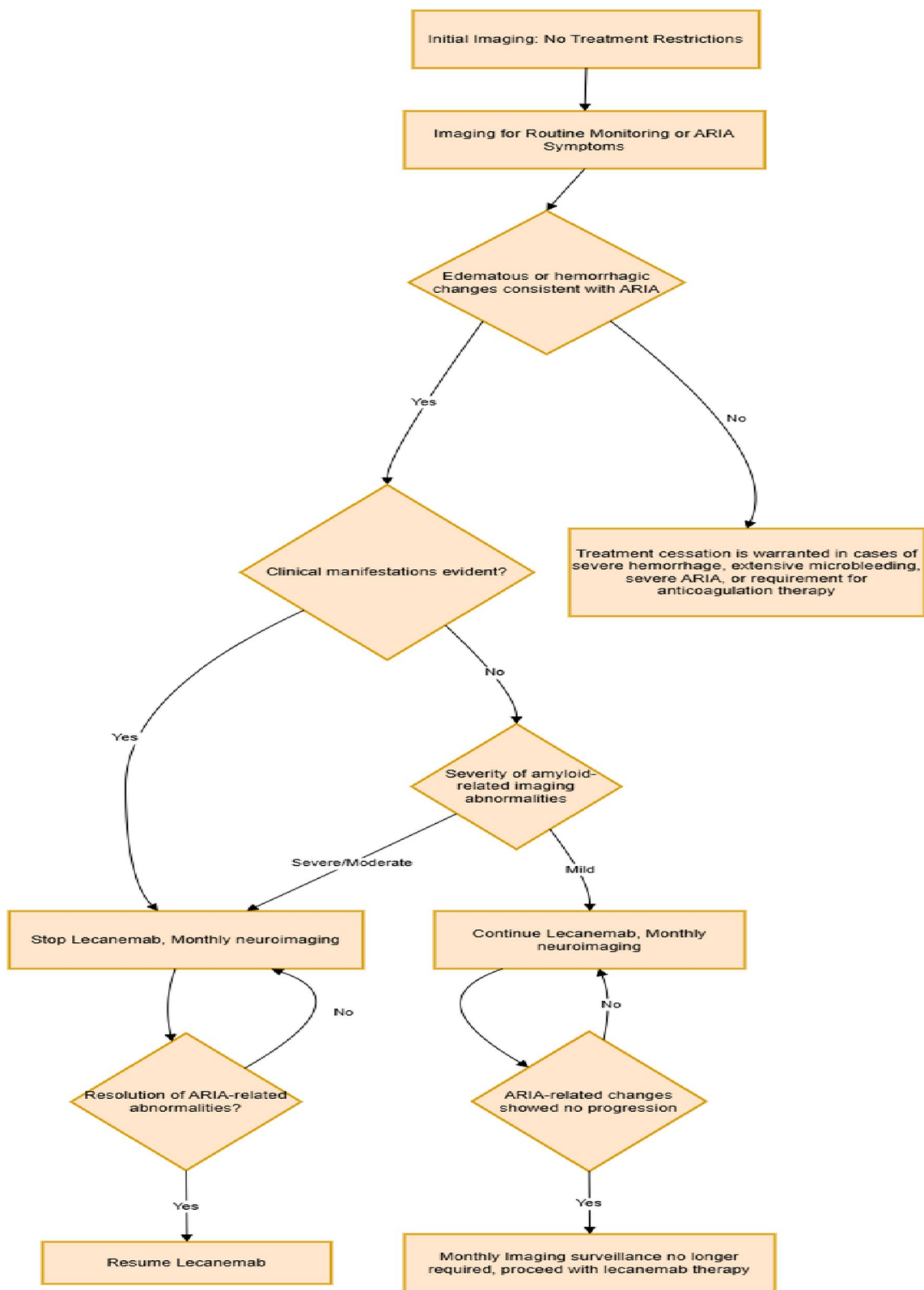


Fig. 2 Imaging-based monitoring and management of ARIA in lecanemab therapy

or result in permanent disability or death [198, 199]. Ongoing research and clinical experience are expected to improve risk stratification for patients predisposed to ARIA events, enhancing personalized treatment strategies.

Beyond amyloid-related imaging abnormalities (ARIA), lecanemab's safety data reveal additional clinically relevant adverse events. In the combined Core and Open-Label Extension trial, non-ARIA effects occurring in at least 10% of participants included infusion-related reactions (nearly one-quarter; 24.5%), COVID-19 (approximately 15%; 14.7%), and headache (roughly 10%; 10.3%) [200]. Infusion-related reactions, predominantly mild-to-moderate (the vast majority; 96.5% of cases), frequently manifested during initial administration (nearly three-quarters; 73% incidence at first infusion), with symptoms such as pyrexia, influenza-like illness, gastrointestinal disturbances, and hemodynamic fluctuations [2]. Recurrence was uncommon, as over two-thirds (65.1%) of affected individuals experienced only one episode [200]. Severe (grade 3–4) reactions were rare, reported in less than 1% (0.6%) of participants exclusively during the first dose [200].

In more severe ARIA-E presentations, temporary treatment discontinuation may be warranted [162]. Available data indicate that ARIA-E typically resolves within 4 months of detection, supporting transient cessation to facilitate radiographic resolution in higher-severity cases [200, 201]. For ARIA-H (hemosiderin deposits), mild radiographic findings often permit uninterrupted dosing, whereas severe presentations may necessitate discontinuation, reflecting a risk-adapted management approach [34, 200, 202]. Infusion-related reactions, while predominantly mild-to-moderate, occasionally escalate to severe (grade 3–4) events. Although such reactions occurred infrequently (0.6% incidence), they may justify treatment termination in exceptional circumstances, though current protocols lack explicit guidance for these

scenarios [200]. This underscores the need for individualized risk–benefit assessments in clinical practice, integrating adverse event severity, patient comorbidities, and biomarker trajectories to optimize therapeutic outcomes.

Table 4 classifies the ARIA-E and ARIA-H severity based on edema size, number of microhemorrhages, and extent of superficial siderosis [201, 202].

Donanemab

Donanemab, a humanized IgG1 antibody, specifically targets the N-terminal pyroglutamate amyloid-beta (A β) epitope exclusively found in brain amyloid plaques, a hallmark of Alzheimer's disease pathology. Designed to facilitate microglial-mediated clearance, donanemab aims to eliminate existing amyloid deposits [203]. In preclinical studies donanemab's target engagement was assessed using the PDAPP mouse model, genetically predisposed to develop A β plaques [204]. Intraperitoneal administration of mE8-IgG2a resulted in a dose-dependent reduction in A β plaque burden without inducing microhemorrhages [204]. Postmortem analysis of brain tissue from Alzheimer's disease (AD) and Down syndrome patients revealed that donanemab bound to approximately one-third of plaques [205]. A phase Ia trial evaluated the safety and efficacy of donanemab in 100 participants diagnosed with mild cognitive impairment (MCI) or mild-to-moderate Alzheimer's disease (AD) dementia, employing a seven-arm design. Single ascending doses (SAD) progressed to multiple ascending doses (MAD), utilizing intravenous doses ranging from 0.3 to 10 mg/kg. The monthly administration of 10 mg/kg resulted in a 40–50% reduction of A β plaques, with ARIA-H observed in two participants and a half-life of approximately 10 days at the highest dosage. Phase Ib involved 61 participants administered 10–40 mg/kg intravenously, showing sustained A β reductions over 72 weeks; ARIA-E was observed in 26% of cases [203, 206].

Table 4 Severity grading of ARIA types

ARIA type [201, 202]	Mild [201, 202]	Moderate[201, 202]	Severe[201, 202]
ARIA-E (edema)	FLAIR signal hyperintensities measuring less than five centimeters in diameter, confined to a solitary sulcus or localized within a discrete cortical or subcortical territory	FLAIR signal hyperintensities measuring between five and ten centimeters in diameter, present at one or multiple locations, with individual lesion diameters not exceeding ten centimeters	Extensive FLAIR signal hyperintensities exceeding 10 cm in diameter, accompanied by gyral expansion and sulcal obliteration; multifocal involvement with multiple discrete lesions is also possible
ARIA-H (microhemorrhage)	Presence of four or fewer novel microhemorrhagic lesions.	Identification of five to nine novel microhemorrhages.	Detection of ten or more newly identified microhemorrhagic lesions
ARIA-H (superficial siderosis)	Isolated focal region of hemosiderin deposition	Identification of two separate localized areas of siderotic change	Extensive siderosis is characterized by three or more focal areas of hemosiderin deposition

The phase II TRAILBLAZER-ALZ trial (NCT03367403) evaluated 272 participants with early symptomatic AD (MMSE 20–28, PET-confirmed A β pathology). Donanemab at 700 mg (initial doses) and 1400 mg (subsequent doses) every 4 weeks for 72 weeks slowed cognitive decline by 25% (iADRS), with 67.8% achieving A β clearance. Significant reductions in plasma biomarkers (p-tau 217, GFAP) correlated with A β levels. ARIA-E was reported in 6.1% of participants [141, 207, 208].

Phase III (TRAILBLAZER-ALZ 2; NCT04437511) expanded criteria to include high tau levels. Among 1800 participants, those with intermediate tau experienced a 35% reduction in cognitive decline (CDR-SB) and 40% improvement in daily activities (iADRS). ARIA-E and ARIA-H were reported in 24% and 31% of participants, respectively. These findings suggest pending FDA approval [209, 210]. Lecanemab's phase III trials reveal primary limitations centered on amyloid-related imaging abnormalities (ARIA), particularly edema (ARIA-E) and microhemorrhages (ARIA-H) [210]. A heightened occurrence of these events in genetically susceptible subgroups, such as APOE ϵ 4 carriers, underscores safety challenges in populations most likely to receive amyloid-targeted therapies. While many ARIA cases were asymptomatic, severe presentations required treatment discontinuation in some participants, complicating sustained therapeutic adherence [210]. The relatively short duration of the data collection period imposes a constraint on the long-term assessment of donanemab's efficacy and safety profile, limiting the comprehension of its sustained effects [210]. Mandatory MRI surveillance for ARIA detection imposes logistical burdens, particularly in resource-limited settings, and may restrict real-world applicability. Trial exclusion criteria likely underestimate ARIA risks in broader clinical populations, raising generalizability concerns. These limitations highlight unresolved safety–efficacy trade-offs and the need for improved risk stratification. Reliance on specialized clinical settings for monitoring risks exacerbating disparities in Alzheimer's care access, emphasizing the necessity for scalable safety protocols and inclusive trial designs [209, 210].

Ongoing phase III studies include TRAILBLAZER-ALZ 3 (NCT05026866; prevention in 3300 participants), TRAILBLAZER-ALZ 4 (NCT05108922; comparison with aducanumab in 200 participants), TRAILBLAZER-ALZ 5 (NCT05508789; efficacy in 1500 participants), and TRAILBLAZER-ALZ 6 (NCT05738486; dosing optimization in 800 participants). These trials aim to further evaluate safety, efficacy, and ARIA-E frequency. However, further long-term and larger-scale trials are essential for a comprehensive evaluation of donanemab's

efficacy and safety in the context of Alzheimer's disease [141].

The therapeutic potential of donanemab in Alzheimer's disease (AD) is constrained in patients with advanced tau pathology, reflecting the complex A β –tau interplay underlying disease progression. Available trial data indicate that donanemab's efficacy correlates with baseline amyloid burden, as individuals with lower initial A β levels demonstrate higher rates of complete amyloid clearance [1, 211, 212]. However, despite robust amyloid reduction, minimal cognitive improvements were observed in cohorts with significant tau accumulation, suggesting that tau-driven neurodegeneration persists as a critical barrier to clinical efficacy [2, 3, 213].

This limitation directly informs donanemab's positioning within the AD treatment paradigm. First, the drug appears optimally effective during early disease stages, prior to widespread tau propagation—a concept aligning with the hypothesized “critical window” wherein A β -targeted interventions may delay downstream tau toxicity [4, 213, 214]. Second, the dissociation between amyloid clearance and cognitive outcomes underscores the necessity for combination therapies concurrently addressing A β and tau pathology to maximize therapeutic impact [4, 215].

Ultimately, donanemab's role may be redefined toward preventive or early-stage treatment, emphasizing timely diagnosis and intervention prior to irreversible tau-mediated damage. This positioning aligns with evolving therapeutic frameworks prioritizing pathogenic sequencing and combinatorial regimens to address AD's multifactorial pathology.

Gantenerumab

A synthetic human combinatorial antibody library produced antibody fragments, later evaluated for their anti-amyloid-beta (A β) effects [216]. Gantenerumab is a fully human monoclonal IgG1 antibody targeting amyloid-beta (A β) that facilitates the removal of amyloid plaques in the brain. It achieves this by promoting dissociation of peptide aggregates and clearing fibrillar A β [217, 218]. The IgG1 backbone engages the Fc gamma receptors, activating microglia to phagocytize aggregated A β [217, 218]. Studies utilizing electron microscopy have confirmed that gantenerumab binds to A β fibrils within plaques in ex vivo samples, while immunofluorescence analysis has demonstrated its binding to A β in the brains of individuals with Alzheimer's disease (AD) [217, 219]. Additionally, live-cell imaging of postmortem AD brain tissue indicates that gantenerumab, after binding to A β plaques, is cleared via Fc gamma receptor-mediated microglial phagocytosis, with subsequent degradation occurring in lysosomes [218]. Gantenerumab exhibits

the highest affinity for aggregated A β , including fibrils and plaques, as well as soluble oligomers. In vitro data suggest that gantenerumab also binds to oligomers, neutralizing their toxicity [217]. This evidence points to two critical actions of gantenerumab on aggregated A β : inhibition of further aggregation and the activation of microglial phagocytosis. Beyond its effects on A β , gantenerumab has been shown to influence several key biomarkers of Alzheimer's disease (AD) pathology and neurodegeneration in clinical trials. These include dose- and time-dependent reductions in cerebrospinal fluid (CSF) levels of total tau (t-tau) and phosphorylated tau (p-tau), as well as decreases in the synaptic biomarker neurogranin [216, 220]. Additionally, reductions in neurofilament light chain (NfL) levels have been observed, further supporting gantenerumab's impact on multiple biological processes underlying AD pathophysiology [216].

Preclinical studies showed its potential to neutralize A β oligomer toxicity and reduce plaque deposition via cell-mediated clearance in transgenic mice, also inhibiting the formation of new plaques [217]. A phase I single ascending dose trial enrolled 18 healthy male participants to assess the safety, tolerability, and pharmacokinetics of subcutaneous gantenerumab. Adverse events (AEs) up to 12 weeks were the primary outcomes, with pharmacokinetic parameters as secondary outcomes, though results are currently unavailable [221]. Another phase I trial involved 60 Alzheimer's disease patients who received 2 to 7 intravenous infusions of gantenerumab (60 or 200 mg) or placebo at 4-week intervals. Dose-dependent reductions in A β were observed using amyloid-PET imaging. Two APOE ϵ 4 homozygous participants in the 200 mg group experienced amyloid-related imaging abnormalities-edema (ARIA-E), prompting subsequent dose adjustments [218]. The SCarlet RoAD trial assessed subcutaneous gantenerumab (105 or 225 mg every four weeks) in prodromal Alzheimer's disease over 2 years, followed by a 2-year open-label extension. The trial failed to show significant differences in primary or secondary endpoints, leading to early termination due to futility. Higher doses and extended titration schedules were suggested to achieve efficacy and manage ARIA-E risk [220]. The DIAN-TU trial examined gantenerumab in individuals with dominantly inherited Alzheimer's disease (DIAD), with participants uptitrated to 1200 mg every 4 weeks over 4 years. ARIA-E occurred in 10 out of 52 participants on active treatment, and ongoing analyses are exploring its distinct effects in DIAD compared to sporadic AD [222, 223].

The GRADUATE I and II trials involved 1966 participants with prodromal to mild Alzheimer's disease,

using a three-step titration to a 1020 mg monthly dose. Despite this approach, the trials failed to achieve significant clinical efficacy, leading to the discontinuation of gantenerumab's development program [224, 225].

The DIAN-TU trial, which evaluated gantenerumab alongside other therapies, targeted a unique cohort with dominantly inherited Alzheimer's disease (DIAD), contrasting with the sporadic AD populations in the Graduate I/II trials [222–225]. In DIAD, gantenerumab induced favorable modulation of soluble CSF biomarkers, including reductions in total tau, p-tau181, and neurogranin, suggesting potential target engagement despite lacking clinical translatability in sporadic AD. This divergence may reflect etiological distinctions: DIAD arises from autosomal dominant mutations driving early A β overproduction, whereas sporadic AD involves multifactorial mechanisms, including APOE ϵ 4-mediated pathways [222–225]. Such genetic heterogeneity implies that anti-A β agents like gantenerumab may exhibit differential efficacy based on a patient's pathogenic drivers. For instance, DIAD's pronounced A β overproduction could enhance responsiveness to amyloid-lowering therapies compared to sporadic cases [222–225].

While these findings carry critical implications for precision medicine frameworks in AD, the DIAN-TU study's limited sample size—a consequence of DIAD's rarity—precludes definitive conclusions [222–225]. Future research must prioritize genetic and biomarker stratification to identify patient subsets most likely to benefit from specific anti-A β therapies. Additionally, optimizing intervention timing relative to disease stage and amyloid burden may enhance therapeutic outcomes across both inherited and sporadic AD populations.

The lack of clinical success in Alzheimer's disease (AD) trials targeting amyloid-beta (A β) may be attributed to several interrelated factors. Inadequate reduction of A β to therapeutic thresholds has been implicated as a key limitation. Early-phase anti-amyloid therapies, such as gantenerumab, employed dosing regimens now recognized as subtherapeutic [226]. For instance, initial doses of 105 mg or 225 mg administered every four weeks in the SCarlet RoAD trial failed to achieve sufficient amyloid clearance, a conclusion supported by subsequent findings from open-label extensions [218, 226]. For these later studies revealed that escalated doses (for example, 1200 mg) correlated with significantly greater A β plaque removal, implying that prior trial designs underestimated the requisite therapeutic threshold for clinical efficacy [227, 228]. This dose-dependent relationship underscores the possibility that earlier interventions lacked the pharmacological intensity needed to modify disease progression meaningfully.

The optimization of therapeutic dosing for anti-amyloid antibodies is a complex challenge, requiring a delicate balance between efficacy and safety. This is particularly pertinent in light of amyloid-related imaging abnormalities (ARIA) associated with higher doses. Furthermore, the failure of clinical trials in Alzheimer's disease (AD) can be attributed, in part, to the inadequate addressing of tau pathology and neuroinflammation, which are downstream consequences of amyloid- β (A β) accumulation. Although anti-amyloid therapies may effectively reduce A β plaques, their impact on secondary pathological processes, including tau-mediated neurodegeneration and neuroinflammation, is often insufficient [216, 226]. The intricate interplay between A β , tau, and neuroinflammation poses a significant obstacle to the development of clinically efficacious treatments. Biomarker results from various trials, including the DIAN-TU-001 study, underscore this challenge. While significant reductions in A β plaques and alterations in tau biomarkers were observed, clinical efficacy was not demonstrated [226]. The timing of intervention in relation to disease progression is another critical factor that warrants consideration in the development of effective therapeutic strategies for AD.

The phase 3 trial outcomes of gantenerumab, a fibril-targeting anti-A β monoclonal antibody, offer pivotal insights into Alzheimer's disease therapeutics. Unlike lecanemab (A β protofibrils) and donanemab (pyroglutamate A β), gantenerumab did not demonstrate

significant slowing of cognitive decline, suggesting that broad amyloid plaque reduction alone may lack clinical translatability [229, 230]. This divergence underscores the importance of selectively targeting pathogenic A β conformers [210, 231]. Lecanemab and donanemab's efficacy implies soluble or modified A β species may contribute more directly to pathology than fibrillar aggregates. Gantenerumab's failure to achieve amyloid-negative thresholds, despite reducing plaque burden, emphasizes the need for near-complete amyloid clearance to confer therapeutic benefit [230].

These findings advocate for precise A β isoform targeting, optimized dosing to maximize amyloid removal and minimize adverse effects, and early intervention prior to irreversible neurodegeneration. Future research should prioritize defining pathologically relevant A β species and therapeutic timing relative to disease stage.

Table 5 shows comparative analysis of anti-amyloid monoclonal antibodies in Alzheimer's disease: mechanisms, clinical efficacy, and safety profiles of lecanemab, donanemab, and gantenerumab.

Discussion

The advancement of immunotherapy in passive form for the treatment of Alzheimer's disease (AD) has attracted considerable interest, especially due to recent clinical trials and their results [232]. Despite the variable outcomes of these trials, it is crucial to

Table 5 Comparison of monoclonal antibodies

Parameters	Lecanemab	Donanemab	Gantenerumab
Target	Soluble A β protofibrils and fibrillar aggregates [132–138]	N-terminal pyroglutamate A β in amyloid plaques [203]	Fibrillar A β , plaques, and soluble oligomers [216, 217, 219]
Mechanism	Neutralizes neurotoxic soluble protofibrils and fibrils; restores neural oscillations [132, 139]	Microglial-mediated clearance of amyloid plaques [203, 204]	Dissociates A β aggregates; activates microglial phagocytosis [217, 218]
Affinity	100x > aducanumab for protofibrils [139]	High specificity for insoluble plaques [141]	High for fibrils/plaques; moderate for oligomers [217]
Clinical efficacy	27% slowing of decline (CDR-SB) at 18 months [146]	25% slowing (iADRS, Phase II) [207, 208]; 35% (CDR-SB, Phase III) [209, 210]	No significant efficacy in Phase 3 [224, 225]
Amyloid reduction	Phase 2: 21.5% CARICs [143]; Phase 3: -70 centiloids [146]	Phase 1: 40–50% plaque reduction [203, 206]; Phase 2: 67.8% clearance [207]	Dose-dependent plaque reduction; subtherapeutic early dosing [218, 226]
ARIA incidence	ARIA-E < 10% [143, 144]; higher in ApoE4 [148, 149]	ARIA-E 6.1% (Phase II) [207, 208]; 24% (Phase III) [209, 210]	ARIA-E 10% (DIAN-TU) [222]; high risk at escalated doses [224, 225]
Dosing	10 mg/kg biweekly [145, 146]	700–1400 mg every 4 weeks [203, 206, 207, 209]	105–1200 mg monthly [218, 220, 224, 225]
Key limitations	Modest effect size; age disparities [146]; ARIA in ApoE4 [148, 149]; requires early diagnosis [151]	Limited efficacy in advanced tau [210, 213]; high ARIA [209, 210]	No clinical benefit in sporadic AD [224, 225]; safety challenges [226, 230]
Unique advantages	Targets soluble protofibrils [132]; lower ARIA vs. others [152]	Rapid plaque clearance; early-stage efficacy [203, 211, 212]	Broad A β targeting; biomarker modulation [216, 220, 222, 223]
Phase 3 outcomes	Met primary endpoints (CLARITY-AD) [146]	Cognitive/functional benefits (TRAILBLAZER-ALZ 2) [209, 210]	Failed (GRADUATE I/II) [224, 225]

acknowledge that setbacks in individual trials do not warrant the discontinuation of passive immunization techniques. These studies offer essential prospects for enhancing future strategies and acquiring vital insights into Alzheimer's disease pathophysiology. In light of our present knowledge of Alzheimer's disease and its treatment options, a thorough assessment of new monoclonal antibodies, namely lecanemab, donanemab, and gantenerumab, is now warranted. These breakthrough therapies have significant potential for tackling the complex issues of Alzheimer's disease, providing renewed optimism for patients and caregivers.

The IgG subclass plays a crucial role in regulating antibody-mediated A β plaque clearance in Alzheimer's disease, especially through its interaction with Fc γ receptors on microglia, which affects the effectiveness of particular subclasses. Research demonstrates that IgG2 antibodies, possessing a superior affinity for these receptors, exhibit enhanced effectiveness in diminishing plaque burden compared to IgG1, as validated by ex vivo experiments using primary mouse microglial cells and prolonged in vivo investigations. Researchers tested six monoclonal antibodies with different IgG subclasses targeting the same N-terminal A β epitope and found that IgG2 antibodies exhibited enhanced plaque clearance and offered more protection against neuritic dystrophy than IgG1. The findings highlight the dual importance of IgG subclass and epitope specificity in maximizing antibody efficacy; antibodies aimed at specific N-terminal A β epitopes exhibited significantly improved therapeutic potential, whereas those targeting C-terminal regions, such as epitope 15–24, were largely ineffective in mitigating amyloid pathology, indicating that specific epitopes are essential for realizing therapeutic advantages [233]. A research demonstrated that antibodies directed against N-terminal sections might significantly diminish neuritic disease, despite their low affinity for soluble A β , implying a new treatment approach: antibodies may confer advantages without interacting with soluble A β . This discovery contests conventional methods by suggesting that focusing on plaque-associated A β may be adequate for diminishing neuronal damage, providing novel insights into the mechanisms of amyloid clearance. These insights underscore the necessity of choosing both the IgG subclass and specific epitope regions to improve therapeutic efficacy in antibody-based treatments for Alzheimer's disease, with N-terminal A β epitopes emerging as particularly promising targets for future therapeutic advancements [233].

Significant advancements have been made in developing therapies for Alzheimer's disease (AD), particularly monoclonal antibodies (mAbs) targeting tau and amyloid-beta (A β). Recent insights into tau's

extracellular behavior and the central role of immune activation in the central nervous system (CNS) have emphasized the importance of antibody design. Current challenges involve the limited availability of extracellular tau compared to intracellular tau. Studies using mass spectrometry reveal that tau proteins in cerebrospinal fluid (CSF) frequently lack N- and C-terminal regions, reducing the targetable population for antibodies directed at these ends. This has shifted focus toward mAbs targeting tau's central region, which may provide a broader range of interaction and improve the ability to inhibit tau seeding and propagation, crucial in addressing tau pathology in neurodegenerative diseases [234–237].

The development of mAbs targeting various A β species involved in AD pathology has shown promising therapeutic and diagnostic outcomes. Lecanemab, which targets protofibrils, has demonstrated a strong association between A β clearance and a reduction in cognitive decline rates. In its phase 2 core study, lecanemab showed rapid A β clearance correlating with clinical benefits in early symptomatic AD. The ongoing AHEAD 3–45 study extends this work by evaluating lecanemab in cognitively normal individuals with intermediate or elevated A β levels, aiming to establish its efficacy in preventing amyloid accumulation, tau pathology progression, and cognitive decline during preclinical AD. This landmark study, designed to assess treatment effects over 216 weeks, will provide critical insights into lecanemab's potential as a preventive therapy for AD [238].

Despite these promising findings, challenges persist. The safety profile of mAbs, particularly with respect to amyloid-related imaging abnormalities (ARIA), necessitates careful consideration. Gantenerumab, for instance, demonstrated potential neuroprotective effects but raised concerns due to the incidence of ARIA in higher-dose groups. This underscores the need for precision in dose optimization to mitigate risks while maximizing therapeutic benefits [239]. Similarly, donanemab, which targets pGlu3-A β , has shown modest success in slowing cognitive and functional decline. Although it demonstrated clinically meaningful benefits in Phase 3 trials, including a 38.6% reduction in disease progression risk, ARIA was a notable adverse effect, especially in carriers of the APOE ϵ 4 allele122, 146, 148.

A critical aspect of advancing these therapies lies in improving patient selection and ensuring equitable representation in clinical trials. The CLARITY-AD trial for lecanemab highlighted disparities in racial and ethnic inclusion, with Black, Hispanic, and Asian populations underrepresented. This is particularly concerning given the documented variability in amyloid positivity and APOE4-associated dementia risks across different

demographics. Tailoring treatment approaches to account for genetic, cultural, and socioeconomic factors is essential for maximizing therapeutic efficacy and accessibility [183, 240, 241].

Donanemab's impact on disease stability is noteworthy, with nearly 47% of treated patients showing no cognitive or functional decline over a year compared to 29% in the placebo group. Such outcomes suggest its potential to extend periods of stability in AD, but larger, long-term studies are necessary to validate these findings and explore its applicability across diverse populations and disease stages [210, 242, 243].

While mAbs like lecanemab and donanemab demonstrate therapeutic promise in AD trials, translating these outcomes to real-world settings faces challenges. First, heterogeneity in patient responses arises from genetic variability, differential A β /tau pathology engagement, and biomarker-defined subgroups with distinct therapeutic vulnerabilities [122, 146, 148]. For example, gantenerumab showed biomarker improvements in DIAN-TU (genetically driven A β overproduction) but lacked clinical efficacy in sporadic AD, underscoring the need for precision in patient stratification. [222–225] Second, cost-effectiveness barriers threaten accessibility: mAbs require specialized infrastructure for intravenous delivery, long-term monitoring (for amyloid angiopathy) and carry high direct costs, disproportionately impacting underrepresented populations with limited healthcare access [183, 240, 241]. Real-world implementation must also address disparities in trial enrollment (for example, underrepresentation of Black/Hispanic populations in CLARITY-AD), which may obscure efficacy and safety profiles across diverse demographics [183, 240, 241].

Balancing optimism with pragmatism, future strategies should prioritize risk-stratified dosing (for example, APOE ϵ 4-guided protocols), cost-reduction initiatives (for example, subcutaneous formulations), and equity-focused trial designs to ensure generalizability. Parallel investments in biomarker-guided combination therapies and non-pharmacological interventions may mitigate limitations of monotherapy approaches.

The evolving field of passive immunization for AD reflects the broader trend toward precision medicine. Advances in antibody engineering and trial design have highlighted the potential of disease-modifying treatments, yet real-world implementation faces significant hurdles. High costs, the complexity of intravenous delivery, and the need for specialized infrastructure remain barriers to widespread adoption. Additionally, the multifaceted nature of AD pathogenesis, influenced by genetic, environmental, and lifestyle factors, necessitates an integrative approach

that combines pharmacological therapies with non-pharmacological interventions.

Future efforts must focus on optimizing combination therapies, refining patient stratification strategies, and developing cost-effective delivery methods to overcome these barriers. The ongoing research into tau and A β -targeting antibodies, such as lecanemab and donanemab, underscores the dynamic nature of AD therapeutics. By addressing these challenges, we move closer to realizing the potential of mAbs as effective, accessible, and personalized treatments for AD, marking a pivotal shift in the landscape of neurodegenerative disease management.

Conclusion

The AD therapeutic landscape is evolving with the emergence of monoclonal antibodies, such as lecanemab, donanemab, and gantenerumab. These agents represent a shift from symptomatic management to disease modification by targeting distinct A β species implicated in neurotoxicity. Their mechanisms—lecanemab's focus on protofibrils, donanemab's specificity for pyroglutamate-modified plaques, and gantenerumab's dual action on A β aggregates and microglial activation—highlight their potential to reshape AD management.

Advances in precision medicine, supported by biomarkers like amyloid and tau PET, enable early diagnosis and personalized treatment strategies. However, challenges remain, including managing amyloid-related imaging abnormalities (ARIA), ensuring diverse trial representation, and addressing therapy accessibility.

Future research should prioritize combinatorial approaches targeting A β , tau, and neuroinflammation while exploring innovations like gene and RNA-based therapies. By advancing our understanding of AD's complex pathophysiology, these therapies hold the promise of transformative disease modification and improved patient outcomes.

Abbreviations

AD	Alzheimer's disease
A β	Amyloid-beta
ARIA	Amyloid-related imaging abnormalities
NFTs	Neurofibrillary tangles
PET	Positron emission tomography
CSF	Cerebrospinal fluid
APP	Amyloid precursor protein
CNS	Central nervous system
mAbs	Monoclonal antibodies
FDA	Food and Drug Administration
GFAP	Glial fibrillary acidic protein

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Author contributions

TBA UA OS and SNK conceptualized the study, SMIA MAA MA and MAS and conducted the literature review and drafting, while editing and supervision were performed by TBA, MAS, MAA, SMIA, MK, SMSA, A.R and MA. All authors have read and agreed to the final version of the manuscript.

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Availability of data and materials

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Declarations

Ethics approval and consent to participate

We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Consent for publication

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Competing interests

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